TITLE

CUSTOMER SEGMENTATION & CHURN PATTERN ANALYTICS IN EUROPEAN BANKING

PROJECT REPORT

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TABLE OF CONTENTS

S.No.	Contents	Page No.
	PRELIMINARY PAGES	
1	Title Page	
2	Table of Contents	
3	Abstract	
	CHAPTER 1: INTRODUCTION	
1.1	Introduction	
1.2	Background of the Study	
1.3	Statement of the Problem	
1.4	Need for the Study	
1.5	Objectives of the Study	
1.6	Research Questions	
1.7	Scope of the Study	
1.8	Research Methodology Overview	
1.9	Significance of the Study	
1.10	Limitations of the Study	
1.11	Chapter Scheme	
	CHAPTER 2: LITERATURE REVIEW	
2.1	Introduction	
2.2	Customer Retention and Customer Churn	

2.3	Customer Relationship Management in Banking	
2.4	Customer Segmentation	
2.5	Data Mining and Customer Churn Analysis	
2.6	Predictive Analytics and Churn Modelling	
2.7	Customer Engagement and Churn	
2.8	Customer Demographics and Financial Characteristics	
2.9	Data Visualization and Business Intelligence	
2.10	Research Gap	
2.11	Summary of Literature	
	CHAPTER 3: RESEARCH METHODOLOGY	
3.1	Introduction	
3.2	Research Design	
3.2.1	Quantitative Research	
3.2.2	Descriptive Research	
3.2.3	Comparative Research	
3.3	Objectives of the Study	
3.4	Data Source	
3.5	Sample Size and Sampling Technique	
3.6	Variables Used in the Study	
3.7	Data Preparation and Cleaning	
3.7.1	Dataset Import	
3.7.2	Column Name Standardization	
3.7.3	Duplicate Record Check	

3.7.4	Missing Value Analysis	
3.7.5	Churn Status Validation	
3.7.6	Variable Conversion	
3.7.7	Customer Segmentation	
3.8	Analytical Tools and Techniques	
3.8.1	Google Colab	
3.8.2	Python	
3.8.3	Pandas	
3.8.4	NumPy	
3.8.5	Matplotlib	
3.8.6	Excel	
3.9	Methods of Data Analysis	
3.9.1	Frequency Analysis	
3.9.2	Percentage Analysis	
3.9.3	Churn Rate Calculation	
3.9.4	Retention Rate Calculation	
3.9.5	Customer Segmentation Analysis	
3.9.6	Comparative Analysis	
3.9.7	Graphical Analysis	
3.10	Ethical Considerations	
3.11	Limitations of the Methodology	
3.12	Chapter Summary	
3.13	Additional Data Quality Assessment	

3.14	Explanation of Key Performance Indicators	
	CHAPTER 4: DATA ANALYSIS, RESULTS AND INTERPRETATION	
4.1	Introduction	
4.2	Overall Customer Churn Analysis	
4.3	Geographical Analysis of Customer Churn	
4.4	Gender-Based Customer Churn Analysis	
4.5	Age-Based Customer Churn Analysis	
4.6	Credit Score Analysis	
4.7	Tenure-Based Customer Churn Analysis	
4.8	Account Balance Analysis	
4.9	Number of Banking Products Analysis	
4.10	Customer Activity Analysis	
4.11	Salary-Based Customer Churn Analysis	
4.12	Comparison of Retained and Exited Customers	
4.13	Year-Wise Analysis	
4.14	Summary of Major Findings	
4.15	Conclusion	
4.16	Identification of High-Churn Customer Segments	
4.17	Comparative Interpretation of Major Customer Segments	
4.18	Limitations in Interpreting High-Churn Segments	
	CHAPTER 5: FINDINGS, SUGGESTIONS AND CONCLUSION	
5.1	Introduction	
5.2	Major Findings of the Study	

5.2.1	Overall Churn	
5.2.2	Geographical Differences	
5.2.3	Age-Related Differences	
5.2.4	Gender Differences	
5.2.5	Product Usage	
5.2.6	Customer Activity	
5.2.7	Account Balance	
5.2.8	Credit Score and Tenure	
5.2.9	Estimated Salary	
5.2.10	Retained and Exited Customer Differences	
5.3	Suggestions and Recommendations	
5.3.1	Investigate High-Churn Geographical Regions	
5.3.2	Develop Appropriate Retention Support for Older Customers	
5.3.3	Improve Customer Engagement	
5.3.4	Review Product Usage and Product Suitability	
5.3.5	Investigate Customer Experience and Satisfaction	
5.3.6	Use Multiple Variables for Customer Risk Assessment	
5.3.7	Monitor Retention Performance	
5.3.8	Maintain Responsible Use of Customer Data	
5.3.9	Segment-Based Retention Prioritization	
5.3.10	Establish a Regular Customer Retention Review	
5.4	Expected Benefits of the Recommendations	
5.5	Managerial Implications	

5.6	Limitations of the Study	
5.7	Scope for Future Research	
5.7.1	Multi-Year Churn Analysis	
5.7.2	Predictive Churn Modelling	
5.7.3	Customer Satisfaction Analysis	
5.7.4	Advanced Customer Segmentation	
5.7.5	Retention Intervention Evaluation	
5.7.6	Explainable Analytics	
5.8	Conclusion	
	CHAPTER 6: CUSTOMER CHURN DASHBOARD DEVELOPMENT	
6.1	Introduction	
6.2	Objectives of the Dashboard	
6.3	Tools and Technologies	
6.4	Dataset Description	
6.5	Proposed Dashboard Architecture	
6.6	Dashboard Layout	
6.6.1	Dashboard Title and Introduction	
6.6.2	Key Performance Indicators	
6.6.3	Customer Churn Distribution	
6.6.4	Geography-Based Analysis	
6.6.5	Age-Based Analysis	
6.6.6	Gender-Based Analysis	
6.6.7	Tenure-Based Analysis	

6.6.8	Activity-Based Analysis	
6.6.9	Product-Based Analysis	
6.6.10	Financial Characteristics	
6.7	Interactive Filters	
6.8	Data Processing and KPI Formulas	
6.9	Dashboard Implementation Procedure	
6.10	Dashboard Testing	
6.11	Dashboard Limitations	
6.12	Future Enhancements	
6.13	Chapter Summary	
6.14	Key Dashboard Performance Indicators	
6.15	Dashboard Interpretation	
6.16	Dashboard Validation Approach	
6.17	Dashboard Limitations and Future Improvements	
	REFERENCES	
	APPENDIX	
A	Dataset Variables and Their Purpose	
B	Analysis Outputs and Supporting Results	
C	Python Code and Project Evidence	

LIST OF TABLES

Table No.	Title of the Table	Page No.
4.1	Overall Customer Churn and Retention	36
4.2	Churn Rate by Geography	37
4.3	Churn Rate by Gender	38
4.4	Churn Rate by Age Group	39
4.5	Churn Rate by Credit Score Group	40
4.6	Churn Rate by Tenure Group	41
4.7	Churn Rate by Balance Group	42
4.8	Churn Rate by Number of Products	44
4.9	Churn Rate by Active Member Status	45
4.10	Churn Rate by Estimated Salary Group	46
4.11	Average Characteristics of Retained and Exited Customers	47
4.12	Year-Wise Churn Summary	48
4.13	Summary of Customer Segments with Higher Observed Churn Rates	50
6.1	Dashboard KPI Summary	74

LIST OF FIGURES

Figure No.	Title of the Figure	Page No.
4.1	Overall Customer Churn Distribution	36
4.2	Customer Churn Rate by Geography	37
4.3	Customer Churn Rate by Gender	38
4.4	Customer Churn Rate by Age Group	39
4.5	Customer Churn Rate by Credit Score Group	41
4.6	Customer Churn Rate by Tenure	42
4.7	Customer Churn Rate by Account Balance Group	43
4.8	Customer Churn Rate by Number of Products	44
4.9	Customer Churn Rate by Account Activity	45
4.10	Customer Churn Rate by Estimated Salary Group	46

ABSTRACT

Title: Customer Segmentation & Churn Pattern Analytics in European Banking

The project titled “Customer Segmentation & Churn Pattern Analytics in European Banking” focuses on analysing customer behaviour, identifying patterns associated with customer attrition, and understanding the factors that distinguish retained customers from exited customers in the European banking sector.

Customer churn is a major concern for banks because losing customers can affect customer relationships, business growth, and long-term profitability. Therefore, understanding customer characteristics and engagement patterns is important for developing effective customer retention strategies.

The study uses a dataset containing 10,000 customer records with details such as geography, gender, age, credit score, tenure, account balance, number of banking products, credit card ownership, active membership, estimated salary, and customer exit status. The analysis focuses on customer segmentation and comparative examination of different customer groups.

The findings show that 2,037 customers exited the bank, representing a churn rate of 20.37%, while 7,963 customers were retained, representing a retention rate of 79.63%. The analysis identifies notable differences in churn across customer segments. Germany recorded a higher churn rate of 32.44% compared with France and Spain. Female customers showed a higher churn rate than male customers. Customers aged 50 and above recorded a churn rate of 45.45%, while inactive members showed a higher churn rate than active members. Product usage analysis also indicated higher churn rates among customers holding three or four banking products; however, these groups have relatively small sample sizes and should be interpreted cautiously.

The project uses descriptive statistical analysis, customer segmentation, comparative analysis, and data visualisation to understand these patterns. The results are presented through tables, charts, and key performance indicators to make the findings easier to interpret.

Based on the analysis, the study recommends that banks strengthen customer engagement, monitor high-churn customer segments, improve customer relationship management, and develop targeted retention strategies. The findings can help banking institutions better understand customer behaviour and support data-driven decision-making.

Overall, this project highlights the importance of customer segmentation and churn analysis in identifying customer attrition patterns and supporting effective customer retention practices in the European banking sector.

CHAPTER I

INTRODUCTION

1.1 Introduction

The banking industry has experienced significant changes due to digital transformation, increasing competition, changing customer expectations, and the availability of customer data. Banks now use data analytics to understand customer behaviour, improve service quality, and support business decision-making.

Customer churn refers to a situation in which customers discontinue their relationship with a bank. Customer churn is an important concern because retaining existing customers can help banks maintain stable relationships, improve customer lifetime value, and reduce the need to continuously acquire new customers.

Customer segmentation is the process of dividing customers into meaningful groups based on common characteristics such as age, geographical location, account balance, credit score, banking activity, and product usage. By combining segmentation with churn analysis, banks can identify customer groups that demonstrate different levels of churn risk.

This study, titled "Customer Segmentation & Churn Pattern Analytics in European Banking," examines customer information to understand churn patterns and identify the characteristics associated with customer exit. The study uses descriptive statistical analysis, customer segmentation, comparative analysis, and data visualization to generate useful business insights.

The project also proposes a dashboard-based approach to present the results in a simple and interactive format. Such a dashboard can help banking professionals monitor churn indicators and explore customer segments more efficiently.

1.2 Background of the Study

Banks collect large amounts of information about customers, including demographic characteristics, account balances, credit scores, transaction-related indicators, product holdings, and engagement levels. Analysing this information can help banks understand differences between customers who remain with the bank and those who leave.

Traditional customer analysis often relies on summary reports. However, data analytics makes it possible to examine several customer characteristics together and identify patterns that may not be immediately visible.

For example, churn rates may differ by geography, age group, gender, product count, or customer activity status. These differences can guide further investigation into customer satisfaction, service quality, product suitability, and customer engagement.

The present study uses a European banking customer dataset containing 10,000 records and 14 variables. The analysis focuses on understanding customer characteristics and comparing churn rates across selected groups.

1.3 Statement of the Problem

Customer churn can affect a bank's customer base, revenue opportunities, and long-term customer relationships. Without systematic analysis, a bank may find it difficult to identify the customer groups with relatively high churn rates or understand which characteristics are associated with customer exit.

The main problem addressed in this study is the need to analyse customer information systematically to identify churn patterns and meaningful customer segments. The study also considers how analytical findings can be presented through visualizations and a dashboard to support managerial decision-making.

1.4 Need for the Study

The study is important for the following reasons:

1. To understand the overall level of customer churn in the selected dataset.
2. To identify customer segments with comparatively high churn rates.
3. To examine differences between customers who exited and customers who remained.
4. To understand the relationship between customer engagement and churn patterns.
5. To provide data-based suggestions for customer retention strategies.
6. To present analytical results through clear charts, tables, and a proposed interactive dashboard.
7. To demonstrate the practical application of data analytics in banking management.

1.5 Objectives of the Study

Primary Objective

To analyse customer segmentation and churn patterns in a European banking dataset using descriptive analytics and data visualization.

Secondary Objectives

1. To calculate the overall customer churn and retention rates.
2. To compare churn rates across geographical, demographic, and behavioural customer segments.
3. To examine churn patterns based on age, gender, tenure, balance, credit score, number of products, and activity status.
4. To compare the characteristics of exited and retained customers.
5. To identify customer groups that may require further investigation for retention purposes.
6. To develop recommendations for improving customer engagement and retention.
7. To design a dashboard concept for presenting customer churn indicators and segment-level insights.

1.6 Research Questions

The study seeks to answer the following questions:

1. What percentage of customers have exited the bank?
2. Which geographical regions show comparatively higher churn rates?
3. How does customer age relate to churn patterns?
4. Do active and inactive customers demonstrate different churn rates?
5. How do product holdings and customer tenure relate to churn?
6. What differences exist between exited and retained customers?
7. How can data visualization help communicate churn patterns to banking decision-makers?

1.7 Scope of the Study

The study focuses on a dataset containing 10,000 customer records associated with European banking. It examines the following areas:

- Customer demographics, including age, gender, and geography.
- Financial characteristics, including credit score, balance, and estimated salary.
- Banking relationship indicators, including tenure, number of products, and credit card ownership.
- Customer engagement, represented by active-member status.
- Customer churn, represented by the exit indicator.
- Data visualization and dashboard-based presentation of results.

The dataset contains records for 2025 only. Therefore, the study focuses on customer-level comparisons and does not establish year-over-year churn trends.

1.8 Research Methodology Overview

The study follows a quantitative, descriptive, and comparative research approach. The customer dataset is analysed using Python and relevant data-analysis libraries.

The methodology includes data inspection, data cleaning, descriptive statistics, customer grouping, churn-rate calculation, comparison of exited and retained customers, and visualization of key findings.

The results are presented through tables and charts. An interactive dashboard may be developed using Streamlit to make the analysis easier to explore.

1.9 Significance of the Study

Significance to Banks

The findings can help banks identify customer groups with relatively high churn rates and prioritize further investigation into customer needs, engagement, and service experiences.

Significance to Management

The analysis provides a structured view of customer behaviour and can support evidence-based discussions about retention initiatives and resource allocation.

Significance to Researchers

The study demonstrates how descriptive analytics can be applied to a real-world business problem using customer-level data.

Significance to Students

The project provides practical experience in data preparation, statistical analysis, visualization, interpretation, and the communication of business insights.

1.10 Limitations of the Study

1. The study uses a single dataset containing 10,000 customer records.
2. The available data covers 2025 only, limiting temporal analysis.
3. The dataset may not include important factors such as customer complaints, service quality, customer satisfaction, or reasons for leaving.
4. The findings describe patterns and associations and do not establish causation.
5. Some customer segments, particularly customers holding three or four products, may have relatively small sample sizes.
6. Estimated salary and account balance values may not fully represent customers' financial circumstances.
7. The results may not be generalizable to every European bank or banking market.
8. A descriptive analysis alone does not provide a validated prediction of future customer churn.

1.11 Chapter Scheme

The report is organized into six chapters:

- Chapter 1 – Introduction: Introduces the research topic, problem, objectives, scope, significance, and limitations.
- Chapter 2 – Review of Literature: Discusses previous research on customer retention, churn analysis, customer segmentation, and analytics.

- Chapter 3 – Research Methodology: Explains the research design, dataset, variables, tools, and analytical methods.
- Chapter 4 – Data Analysis, Results and Interpretation: Presents the findings through tables, charts, and interpretations.
- Chapter 5 – Findings, Suggestions and Conclusion: Summarizes the major findings and presents recommendations and conclusions.
- Chapter 6 – Dashboard Development: Describes the design, components, implementation approach, and testing of the customer churn dashboard.

CHAPTER II

LITERATURE REVIEW

2.1 Introduction

A review of literature provides a foundation for understanding the concepts, methods, and previous findings related to customer churn and segmentation. It helps identify commonly used analytical approaches and explains how the present study contributes to the topic.

Customer churn analysis has received attention in marketing, banking, telecommunications, retail, and other service industries. Researchers have examined customer retention, relationship management, predictive modelling, data mining, and customer behaviour to understand why customers discontinue services.

This chapter reviews important research and concepts relevant to customer segmentation and churn analytics.

2.2 Customer Retention and Customer Churn

Customer retention refers to the ability of an organization to maintain its relationships with existing customers. Customer churn represents the loss of customers over a specified period or within a defined dataset.

Reichheld and Sasser (1990) discussed the economic importance of customer retention and highlighted the potential benefits of maintaining long-term customer relationships. Their work helped establish customer retention as an important managerial concern.

In banking, customer retention may be influenced by service experience, trust, product suitability, convenience, pricing, customer support, and competing offers. However, the importance of these factors can differ across customer groups.

The present study examines churn using the dataset's exit indicator. It compares exited and retained customers to identify observable differences in their characteristics.

2.3 Customer Relationship Management in Banking

Customer relationship management involves understanding customer needs and managing interactions throughout the customer relationship. Banks can use customer information to improve communication, personalize services, and identify opportunities to strengthen customer relationships.

Kumar and Reinartz (2018) discussed customer relationship management as a strategic approach that connects customer information with decisions intended to create value for customers and organizations.

Customer analytics can support these activities by identifying groups with different behavioural patterns. For example, inactive customers may require engagement initiatives, while customers with changing product usage may require further assessment.

The current study contributes to this area by comparing churn rates across customer activity groups and other available customer characteristics.

2.4 Customer Segmentation

Customer segmentation divides a customer population into groups based on shared characteristics. Segmentation may be based on demographic, geographic, financial, behavioural, or transactional information.

In banking, segmentation can help institutions design more relevant services and understand the needs of different customer groups. Examples include age-based segments, geographical segments, active and inactive customers, and groups based on the number of banking products held.

Segmentation may be performed using predefined categories or statistical techniques such as clustering. The selection of an appropriate method depends on the research objective and the available data.

In this study, customers are grouped using predefined categories such as age, geography, tenure, balance, credit score, and activity status. These groupings enable direct comparison of churn rates.

2.5 Data Mining and Customer Churn Analysis

Data mining involves discovering useful patterns in large datasets using statistical and computational techniques. It can be used for classification, clustering, prediction, and exploratory analysis.

Han, Kamber, and Pei (2012) described data mining methods used to extract useful patterns and knowledge from data. Such techniques provide a foundation for customer analytics.

Churn analysis may use descriptive statistics to explain existing patterns or predictive models to estimate the probability that a customer will leave. Common predictive methods include logistic regression, decision trees, random forests, and gradient-boosting algorithms.

The present study emphasizes descriptive and comparative analysis. Predictive modelling would require additional development, validation, and performance assessment before the results could be used for operational decisions.

2.6 Predictive Analytics and Churn Modelling

Predictive analytics uses historical data and statistical or machine-learning techniques to estimate future outcomes.

Neslin et al. (2006) examined customer churn management in the context of the telecommunications industry and discussed the role of churn prediction and management. Although their research focuses on telecommunications, the broader concepts are relevant to customer-retention problems in other service industries.

Verbeke et al. (2012) investigated customer churn prediction using data-mining techniques and emphasized the importance of identifying customers who are more likely to leave.

These studies demonstrate that predictive models can help organizations prioritize customer-retention activities. However, the usefulness of a model depends on data quality, appropriate evaluation methods, and the business consequences of false predictions.

The current project does not treat descriptive churn rates as individual predictions. It identifies groups with higher observed churn rates and provides a basis for further analysis.

2.7 Customer Engagement and Churn

Customer engagement refers to the extent to which customers interact with or actively use an organization's services. In banking, engagement may include account usage, digital banking activity, transactions, and interactions with customer support.

Customers who use banking services regularly may demonstrate different retention patterns from customers who are less active. Nevertheless, inactivity alone does not prove that a customer will leave.

The present study examines the dataset's active-member indicator and compares churn rates between active and inactive customers. Additional information, such as transaction frequency and digital-channel usage, could improve future research.

2.8 Customer Demographics and Financial Characteristics

Demographic characteristics such as age, gender, and geography can help researchers understand differences in customer behaviour. Financial characteristics such as account balance, credit score, and estimated salary can also be examined.

These variables should be interpreted carefully. A difference in churn rates between two groups does not necessarily mean that the characteristic itself causes customer exit.

This study compares churn rates across demographic and financial segments to identify patterns that may require further investigation. It does not assume that any demographic characteristic independently explains churn.

2.9 Data Visualization and Business Intelligence

Data visualization presents data through charts, graphs, and dashboards. It helps users identify differences, compare groups, and communicate findings.

Dashboards can combine key performance indicators, filters, charts, and summary tables in a single interface. In customer churn analysis, a dashboard may display overall churn rate, retention rate, geographical comparisons, age-group analysis, and activity-based comparisons.

The project proposes using Python-based visualization and a Streamlit dashboard to make the findings accessible to users who may not be familiar with programming.

2.10 Research Gap

The reviewed literature demonstrates the importance of customer retention, segmentation, data mining, and churn prediction. However, organizations also require clear descriptive analysis that explains how observed churn patterns differ across customer groups.

For the dataset used in this project, the study focuses on:

1. Combining customer segmentation with descriptive churn analysis.
2. Comparing demographic, geographical, financial, and engagement-related characteristics.

3. Presenting the findings through structured tables and visualizations.
4. Translating analytical findings into practical retention recommendations.
5. Organizing the results in a dashboard format for easier exploration.

The project is limited to the available dataset and does not claim that its findings represent every European banking institution.

2.11 Summary of Literature

The literature indicates that customer retention is an important business objective, segmentation can help organizations understand customer groups, and data-mining techniques can support churn analysis and prediction.

The studies reviewed also highlight the importance of appropriate analytical methods and careful interpretation. Building on these concepts, the present project uses descriptive analytics to compare churn patterns across selected customer segments and proposes a dashboard to communicate the results.

CHAPTER III

RESEARCH METHODOLOGY

3.1 Introduction

Research methodology explains the systematic procedures adopted to conduct a study, collect and process data, apply analytical techniques, and interpret the results. It provides a structured approach to achieving the research objectives and ensuring that the findings are meaningful and reliable.

The present study, titled "Customer Segmentation & Churn Pattern Analytics in European Banking," focuses on understanding customer attrition patterns in a European banking dataset. Customer churn refers to the situation in which a customer leaves a bank or discontinues the banking relationship.

Customer churn is an important concern for banks because losing existing customers can affect revenue, customer relationships, and long-term business growth. By identifying customer groups with relatively high churn rates, banks can develop more focused retention strategies.

This study uses Python-based data analysis to examine customer characteristics such as age, geography, gender, credit score, account balance, tenure, number of products held, credit card ownership, activity status, and estimated salary. The analysis is performed using Google Colab, Pandas, NumPy, and Matplotlib.

3.2 Research Design

The study adopts a quantitative, descriptive, and comparative research design.

3.2.1 Quantitative Research

Quantitative research involves analyzing numerical data to identify patterns, differences, and relationships. In this study, customer records are examined using customer counts, percentages, averages, and churn rates.

3.2.2 Descriptive Research

Descriptive research is used to understand the characteristics of the dataset and describe customer churn patterns. It helps identify the proportion of customers who exited and the proportion who remained with the bank.

3.2.3 Comparative Research

Comparative analysis is used to examine differences between customer groups. For example, the study compares churn rates among geographical regions, age groups, gender categories, product groups, and active and inactive customers.

The research design is appropriate because the study aims to understand existing patterns rather than establish cause-and-effect relationships.

3.3 Objectives of the Study

The major objective is to analyze customer segmentation and churn patterns in a European banking dataset.

The specific objectives are:

1. To measure the overall customer churn and retention rates.
2. To analyze customer churn across different geographical regions and demographic groups.
3. To examine the relationship between customer age, credit score, account balance, tenure, and churn status.
4. To compare churn rates according to the number of banking products held by customers.
5. To evaluate differences in churn between active and inactive customers.
6. To compare the average characteristics of exited and retained customers.
7. To suggest customer retention strategies based on the observed analytical findings.

3.4 Data Source

The study uses a structured CSV dataset containing European banking customer information. The dataset was uploaded to Google Colab and analyzed using Python.

The dataset contains 10,000 customer records and 14 columns. It includes demographic information, financial characteristics, customer engagement variables, and a customer exit indicator.

The data includes a year field, which contains only the year 2025. Therefore, the study examines customer churn patterns within the available dataset rather than changes across multiple years.

The analysis is based on the uploaded dataset and should not automatically be interpreted as a complete representation of every European bank or all European banking customers.

3.5 Sample Size and Sampling Technique

The dataset contains 10,000 customer records. All valid customer records available in the dataset were included in the analysis after checking for duplicate rows and valid churn-status values.

The study follows a dataset-based, non-probability approach because the researcher analyzes the available records rather than selecting a new random sample from the entire European banking population.

The sample size for the analysis is 10,000 customer records.

3.6 Variables Used in the Study

The following variables were examined.

Variable	**Description**	**Analytical purpose**
CustomerId	Unique customer identifier	Record identification
Year	Year associated with the record	Dataset period
Geography	Customer's geographical category	Regional comparison
Gender	Customer gender category	Demographic comparison
Age	Customer age	Age-group analysis
CreditScore	Customer credit score	Credit-score segmentation
Tenure	Length of the banking relationship	Relationship-duration analysis

Balance	Customer account balance	Financial segmentation
NumOfProducts	Number of banking products held	Product-usage analysis
HasCrCard	Credit card ownership indicator	Product ownership comparison
IsActiveMember	Customer activity indicator	Engagement analysis
EstimatedSalary	Estimated customer salary	Income-related segmentation
Exited	Customer exit indicator	Churn measurement
Surname	Customer surname	Dataset identification field; not used as a predictive factor

The Exited variable is the primary outcome variable. A value of 1 represents an exited customer, while a value of 0 represents a retained customer.

3.7 Data Preparation and Cleaning

Data preparation was performed before calculating the results.

3.7.1 Dataset Import

The CSV file was uploaded into Google Colab and imported into a Pandas DataFrame. The dataset structure, number of rows, number of columns, and available variables were inspected.

3.7.2 Column Name Standardization

Column names were stripped of unnecessary leading and trailing spaces to support consistent identification of variables.

3.7.3 Duplicate Record Check

The dataset was checked for exact duplicate rows. Exact duplicates were removed before the main analysis to reduce the risk of counting identical records more than once.

3.7.4 Missing Value Analysis

Missing values were checked for every column. The data-quality report generated from the uploaded dataset showed zero missing values across the 14 columns.

3.7.5 Churn Status Validation

The churn variable was converted to a numerical format, and only records with valid values of 0 or 1 were retained for analysis.

3.7.6 Variable Conversion

Relevant numerical fields, including age, credit score, balance, tenure, number of products, activity status, salary, and year, were converted to numerical data types where required.

3.7.7 Customer Segmentation

Customer groups were created for analysis. Age was divided into four categories: under 30, 30–39, 40–49, and 50 and above. Tenure was grouped into 0–2 years, 3–5 years, and 6 or more years. Credit scores and salary values were divided into quartile-based groups. Account balances were analyzed using a zero-balance category and positive-balance groups.

These groupings support comparisons between customers with different characteristics.

3.8 Analytical Tools and Techniques

3.8.1 Google Colab

Google Colab was used as the cloud-based Python environment for executing the analysis without requiring a local Python installation.

3.8.2 Python

Python was used to perform data preparation, calculations, segmentation, and result generation.

3.8.3 Pandas

Pandas was used to import the CSV file, inspect the dataset, calculate summary tables, group customer records, and export results to Excel.

3.8.4 NumPy

NumPy was used for numerical operations and the creation of customer segmentation intervals.

3.8.5 Matplotlib

Matplotlib was used to generate charts showing customer churn distribution and churn rates across selected customer groups.

3.8.6 Excel

The analysis results were exported to an Excel workbook for reviewing the findings, preparing report tables, and supporting further interpretation.

3.9 Methods of Data Analysis

The study uses the following analytical methods.

3.9.1 Frequency Analysis

Frequency analysis measures the number of customers in each category, including customers who exited and customers who remained.

3.9.2 Percentage Analysis

Percentage analysis expresses customer counts as a proportion of the total number of customers.

3.9.3 Churn Rate Calculation

Customer churn rate is calculated using the formula:

Churn Rate (%) = (Number of Exited Customers / Total Customers in the Group) × 100

For example, the overall churn rate is calculated by dividing 2,037 exited customers by 10,000 total customers and multiplying by 100.

3.9.4 Retention Rate Calculation

Customer retention rate is calculated as:

Retention Rate (%) = (Number of Retained Customers / Total Customers) × 100

3.9.5 Customer Segmentation Analysis

Customer segmentation is used to compare churn rates across geographical regions, age groups, gender, credit scores, tenure, account balances, product counts, activity status, and salary groups.

3.9.6 Comparative Analysis

The average numerical characteristics of exited and retained customers are compared to identify differences between the two groups.

3.9.7 Graphical Analysis

Bar charts and other visual representations are used to present the results clearly and identify patterns that may not be immediately visible in raw data tables.

3.10 Ethical Considerations

The study uses an existing structured dataset rather than collecting new personal information directly from customers. The analysis is conducted for academic purposes.

Customer identifiers and surnames are not required for interpreting churn patterns and should not be included in publicly presented findings. Any future use of actual bank customer data should follow applicable privacy requirements, organizational permissions, and data-protection policies.

The results are presented as descriptive findings. Differences between customer groups are not treated as proof that a particular characteristic causes customer churn.

3.11 Limitations of the Methodology

The study has the following limitations:

1. The analysis depends on the completeness and accuracy of the available dataset.
2. The dataset contains records for 2025 only, limiting year-over-year analysis.
3. The analysis is primarily descriptive and does not establish causal relationships.
4. The dataset may not represent every European bank or all banking customers.
5. Important factors such as service quality, customer complaints, competitor offers, and reasons for account closure are not included in the available variables.

6. Small customer groups, particularly those holding three or four products, require cautious interpretation.
7. The study does not yet use a validated predictive model to estimate future individual customer churn.

3.12 Chapter Summary

This chapter explained the research methodology adopted for analyzing customer segmentation and churn patterns in European banking. It described the research design, objectives, data source, sample size, variables, data-cleaning procedures, analytical tools, and statistical methods.

The next chapter presents the actual results obtained from the dataset and interprets the differences in churn rates across customer groups.

3.13 Additional Data Quality Assessment

The quality of the dataset is important for producing reliable analytical results. Before conducting the customer segmentation analysis, the dataset was examined to understand its structure, completeness, and consistency.

The dataset contains 10,000 customer records and 14 variables. The data-quality assessment included checking missing values, duplicate records, customer identifiers, variable formats, and the customer exit indicator.

The data-quality report generated during the analysis showed zero missing values across the 14 columns. The CustomerId field contained 10,000 unique customer identifiers. The dataset included three geographical categories, namely France, Germany, and Spain, and two gender categories.

The Exited variable was used as the main outcome variable. Customers with an Exited value of 1 were classified as exited, while customers with a value of 0 were classified as retained.

These checks helped establish the dataset's suitability for the descriptive and comparative analysis performed in this project.

3.14 Explanation of Key Performance Indicators

The following indicators were used to evaluate customer churn and retention.

1. Total Customer Count

The total number of customer records included in the analysis.

2. Exited Customers

The number of customers whose Exited value is 1.

3. Retained Customers

The number of customers whose Exited value is 0.

4. Customer Churn Rate

Customer churn rate represents the percentage of customers who have exited.

Formula:

Churn Rate = (Exited Customers / Total Customers) × 100

5. Customer Retention Rate

Customer retention rate represents the percentage of customers who remain with the bank.

Formula:

Retention Rate = (Retained Customers / Total Customers) × 100

6. Segment-Wise Churn Rate

Segment-wise churn rate measures the percentage of exited customers within a particular customer group.

Formula:

Segment Churn Rate = (Exited Customers in the Segment / Total Customers in the Segment) × 100

These indicators provide a consistent method for comparing churn patterns across different customer groups.

CHAPTER IV

DATA ANALYSIS, RESULTS AND INTERPRETATION

4.1 Introduction

This chapter presents the results of the analysis conducted on the European banking customer dataset. The analysis focuses on overall churn, customer retention, geographical differences, demographic characteristics, financial variables, product usage, and customer activity.

The dataset contains 10,000 customer records. The results were generated using Python in Google Colab and exported to Excel for further examination.

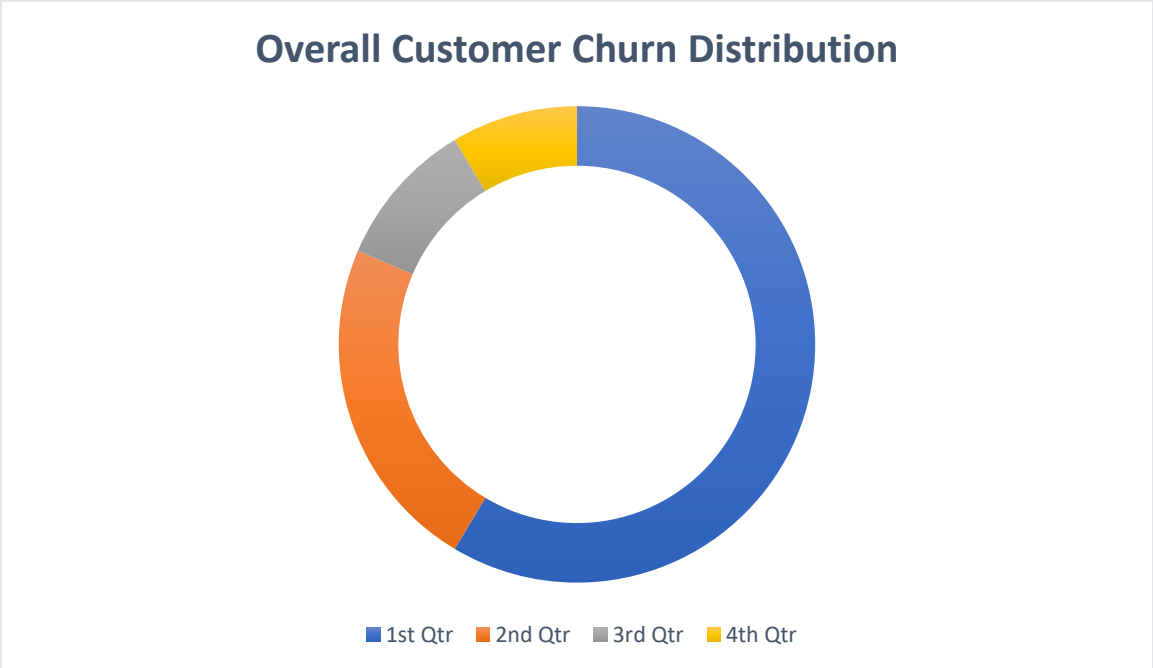
The main purpose of this chapter is to identify customer groups with relatively high churn rates and interpret how their characteristics differ from those of retained customers.

4.2 Overall Customer Churn Analysis

Table 4.1: Overall Customer Churn and Retention

Particulars	Number of customers	Percentage
Total customers	10,000	100.00%
Exited customers	2,037	20.37%
Retained customers	7,963	79.63%

Figure 4.1: Overall Customer Churn Distribution



Interpretation:

The analysis indicates that the dataset contains 10,000 customers, of whom 2,037 have exited the bank. This represents a churn rate of 20.37%. The remaining 7,963 customers, equivalent to 79.63%, are classified as retained.

The results show that approximately one in five customers in the dataset has exited. Although most customers remain with the bank, the proportion of exited customers is substantial enough to justify further examination of churn patterns.

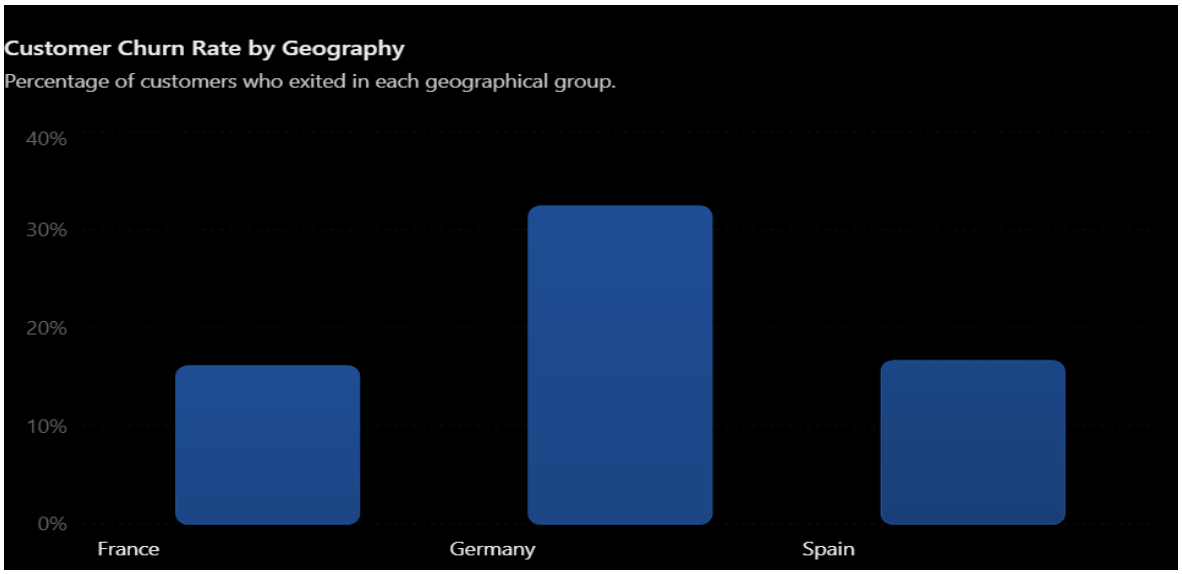
The overall churn rate provides a baseline for comparing the performance of different customer segments.

4.3 Geographical Analysis of Customer Churn

Table 4.2: Churn Rate by Geography

Geography	Customers	Exited	Retained	Churn rate
France	5,014	810	4,204	16.15%
Germany	2,509	814	1,695	32.44%
Spain	2,477	413	2,064	16.67%
Total	10,000	2,037	7,963	20.37%

Figure 4.2: Customer Churn Rate by Geography



Interpretation:

Germany records the highest churn rate at 32.44%, followed by Spain at 16.67% and France at 16.15%.

Although Germany accounts for fewer total customers than France, it has a similar number of exited customers. This indicates that churn is disproportionately high among German customers in the dataset.

The bank should investigate whether differences in service experience, customer expectations, product suitability, pricing, or competition may be associated with this pattern. These possible explanations require additional evidence and cannot be confirmed from the current dataset alone.

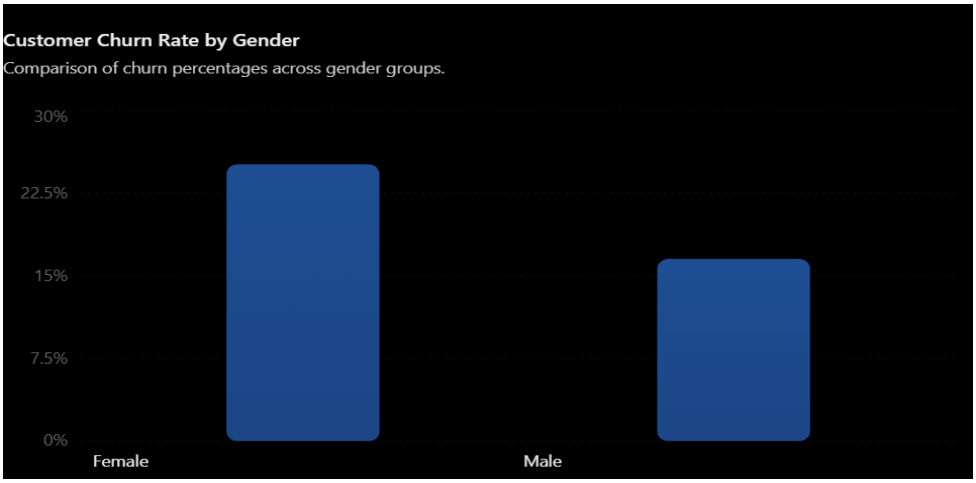
France and Spain have substantially lower churn rates than Germany, suggesting that the bank should examine what distinguishes customer experiences in these regions.

4.4 Gender-Based Customer Churn Analysis

Table 4.3: Churn Rate by Gender

Gender	Customers	Exited	Retained	Churn rate
Female	4,543	1,139	3,404	25.07%
Male	5,457	898	4,559	16.46%

Figure 4.3: Customer Churn Rate by Gender



Interpretation:

The churn rate among female customers is 25.07%, compared with 16.46% among male customers. The difference is 8.61 percentage points.

The result indicates that female customers in this dataset have a higher observed churn rate. However, gender alone should not be treated as the reason for customer exit.

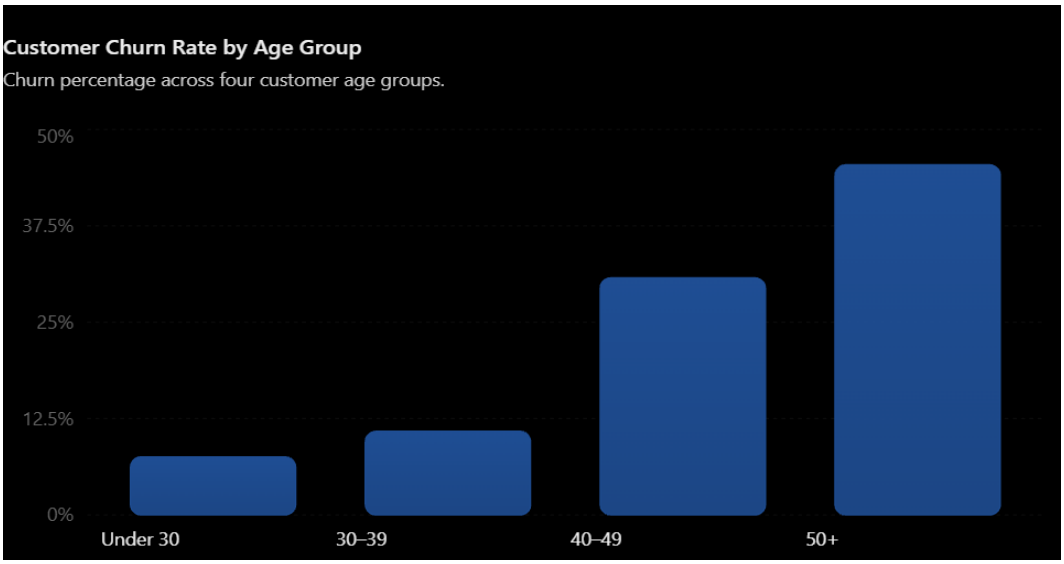
The bank should investigate whether differences in customer age, product usage, account activity, service needs, or other characteristics may help explain the observed gap. Retention initiatives should be based on relevant customer needs and validated evidence rather than assumptions about gender.

4.5 Age-Based Customer Churn Analysis

Table 4.4: Churn Rate by Age Group

Age group	Customers	Exited	Retained	Churn rate
Under 30	1,641	124	1,517	7.56%
30–39	4,346	473	3,873	10.88%
40–49	2,618	806	1,812	30.79%
50 and above	1,395	634	761	45.45%

Figure 4.4: Customer Churn Rate by Age Group



Interpretation:

The age-group analysis reveals substantial differences in customer churn. Customers under 30 have the lowest churn rate at 7.56%, while customers aged 50 and above have the highest rate at 45.45%.

The churn rate rises noticeably among customers aged 40 and above. The 40–49 age group records a churn rate of 30.79%, while almost half of the customers aged 50 and above have exited.

This pattern makes older customers an important segment for further investigation. The bank could examine whether these customers have different expectations concerning service quality, product suitability, accessibility, fees, or support.

The findings do not establish that age causes churn. Further analysis should determine whether age remains associated with exit status after considering other customer characteristics.

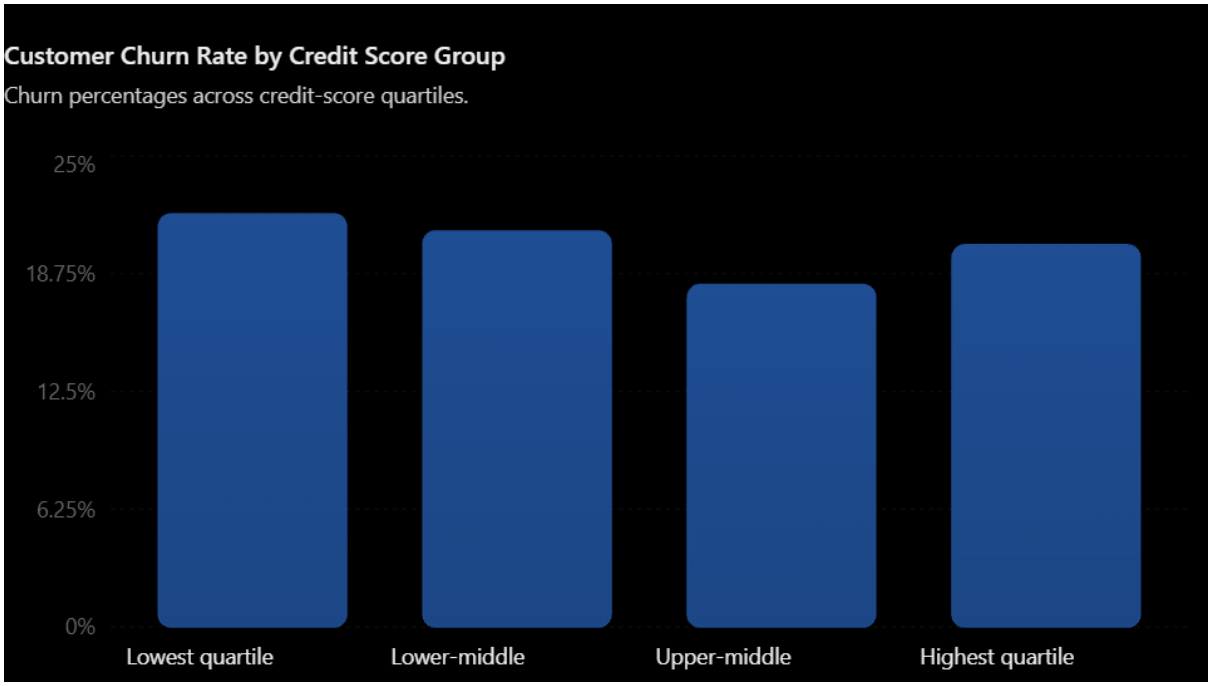
4.6 Credit Score Analysis

Table 4.5: Churn Rate by Credit Score Group

Credit score group	Customers	Exited	Retained	Churn rate
350–584 approximately	2,534	556	1,978	21.94%
Above 584–652	2,501	526	1,975	21.03%
Above 652–718	2,502	455	2,047	18.19%
Above 718–850	2,463	500	1,963	20.30%

Note: The groups are based on quartiles calculated from the dataset. The displayed boundaries are rounded for presentation.

Figure 4.5: Customer Churn Rate by Credit Score Group



Interpretation:

The lowest credit-score group has a churn rate of 21.94%, while the group with credit scores between approximately 652 and 718 has the lowest rate at 18.19%.

The highest credit-score group records a churn rate of 20.30%. Therefore, the relationship between credit score and churn is not consistently increasing or decreasing across the groups.

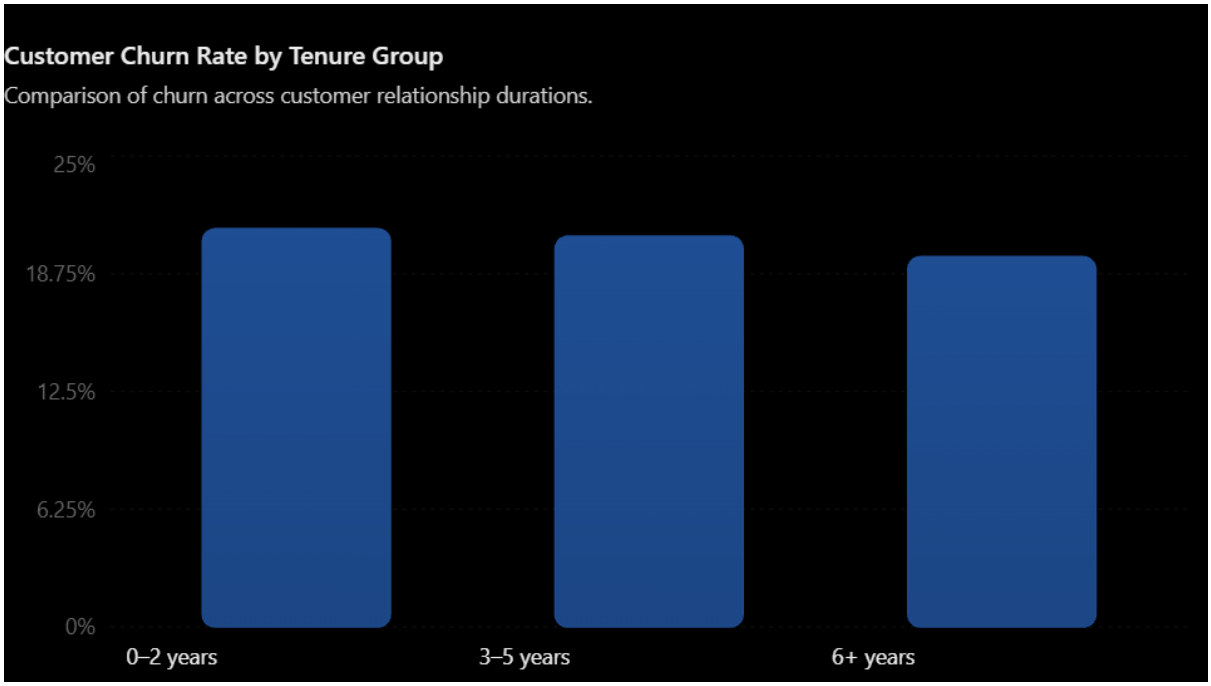
Credit score alone does not appear to provide a clear explanation of the observed churn pattern. The bank should consider other characteristics, such as age, account activity, product usage, and customer experience, when investigating churn.

4.7 Tenure-Based Customer Churn Analysis

Table 4.6: Churn Rate by Tenure Group

Tenure group	Customers	Exited	Retained	Churn rate
0–2 years	2,496	528	1,968	21.15%
3–5 years	3,010	625	2,385	20.76%
6 years and above	4,494	884	3,610	19.67%

Figure 4.6: Customer Churn Rate by Tenure



Interpretation:

The churn rate ranges from 19.67% to 21.15% across the three tenure groups. Customers with a banking relationship of 0–2 years have the highest churn rate at 21.15%, while customers with six or more years of tenure have the lowest rate at 19.67%.

The difference is relatively small compared with the differences observed across age groups and geographical regions.

The findings suggest that tenure alone does not distinguish high-churn customers as strongly as some other variables in this dataset. Nevertheless, banks may still benefit from monitoring customer engagement during the early stages of the relationship.

4.8 Account Balance Analysis

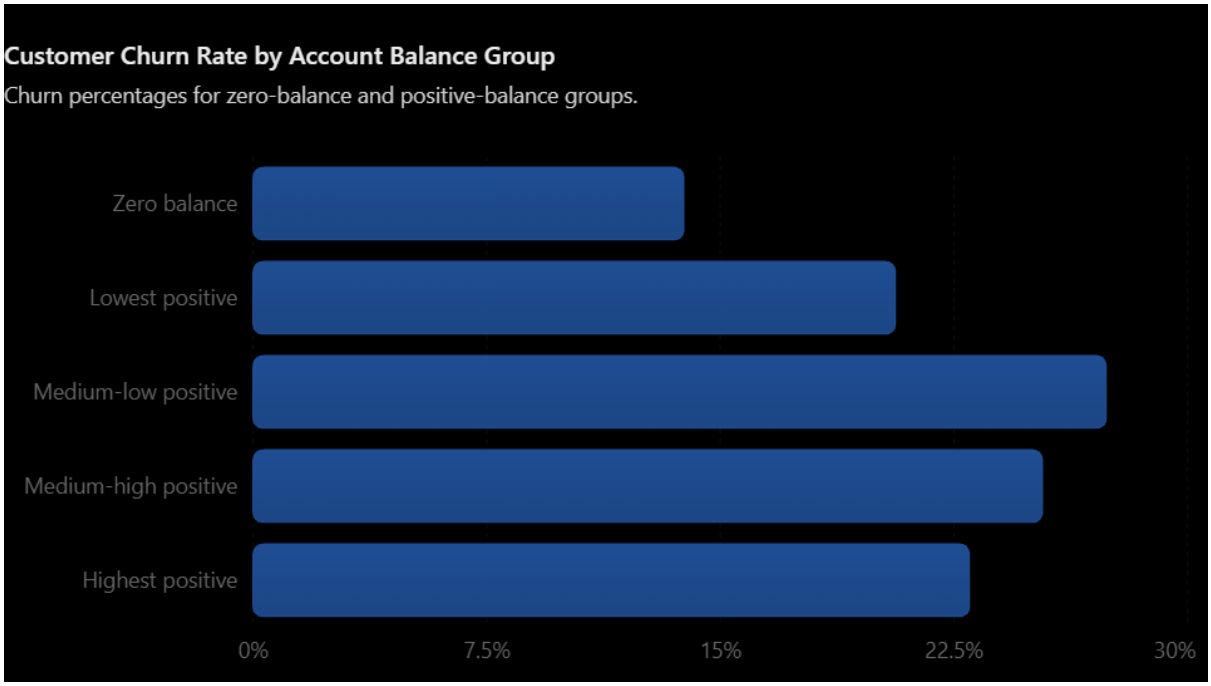
Table 4.7: Churn Rate by Balance Group

Balance group	Customers	Exited	Retained	Churn rate
Zero balance	3,617	500	3,117	13.82%
Positive balance, lowest quartile	1,596	329	1,267	20.61%
Positive balance, second quartile	1,596	437	1,159	27.38%

Positive balance, third quartile	1,595	404	1,191	25.33%
Positive balance, highest quartile	1,596	367	1,229	22.99%

Note: Positive-balance groups use the quartile boundaries generated by the analysis. The zero-balance group is reported separately.

Figure 4.7: Customer Churn Rate by Account Balance Group



Interpretation:

Customers with zero account balances have a churn rate of 13.82%, which is lower than the churn rates observed in the positive-balance groups.

Among customers with positive balances, the second balance quartile has the highest churn rate at 27.38%. The highest-balance quartile records a churn rate of 22.99%.

The relationship is not consistently increasing with account balance. Therefore, higher balances alone cannot be treated as a reliable indicator of churn.

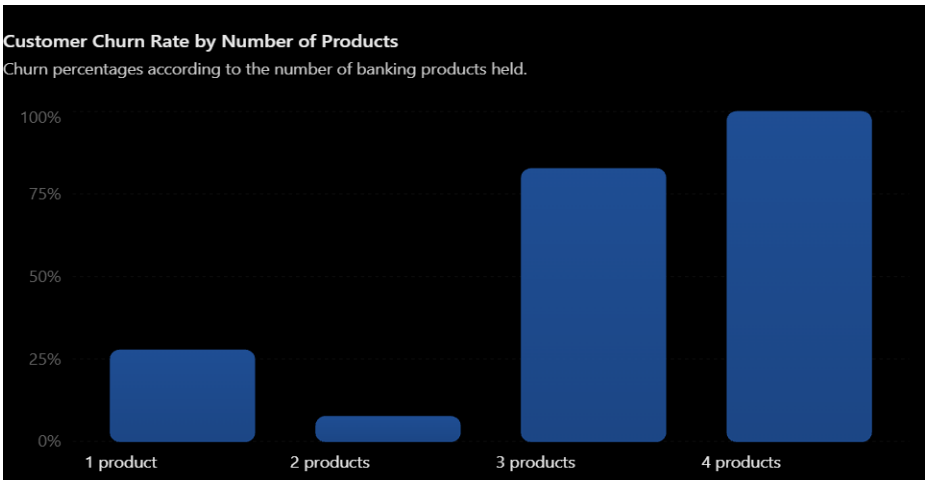
The bank should investigate the relationship between balances, customer activity, age, product usage, and exit status before deciding which financial segments require additional retention attention.

4.9 Number of Banking Products Analysis

Table 4.8: Churn Rate by Number of Products

Number of products	Customers	Exited	Retained	Churn rate
1	5,084	1,409	3,675	27.71%
2	4,590	348	4,242	7.58%
3	266	220	46	82.71%
4	60	60	0	100.00%

Figure 4.8: Customer Churn Rate by Number of Products



Interpretation:

The product-count analysis shows substantial differences in churn rates. Customers holding two products have a churn rate of 7.58%, compared with 27.71% among customers holding one product.

Customers holding three products have a churn rate of 82.71%, while all 60 customers holding four products are recorded as exited.

The high rates among customers with three and four products require cautious interpretation because these groups contain only 266 and 60 customers, respectively. The four-product group is particularly small.

The findings suggest that product count is associated with observed churn patterns in this dataset, but they do not establish why customers with three or four products exited. The bank should investigate whether these groups reflect unusual product combinations, customer circumstances, or data characteristics.

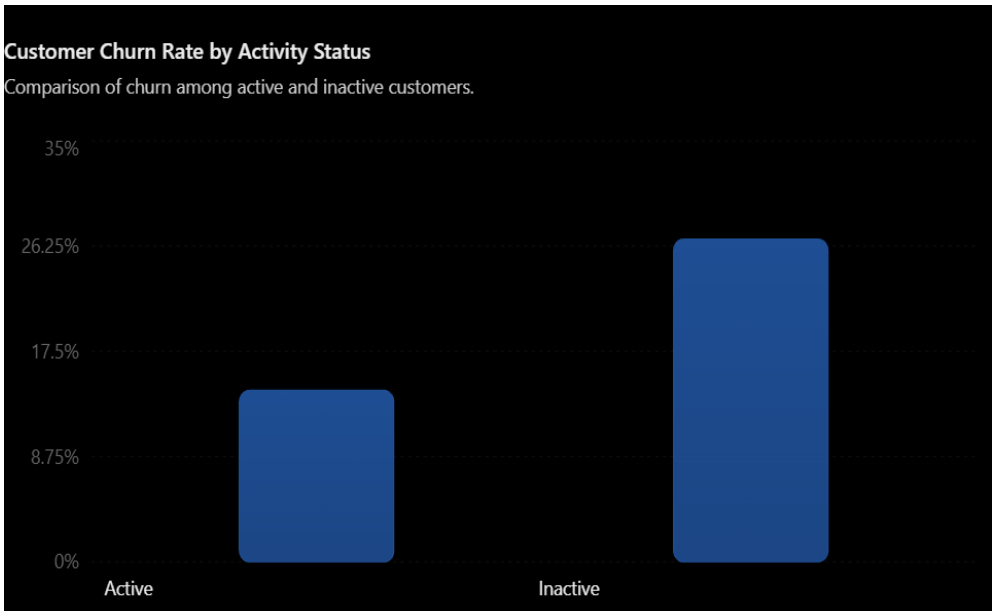
The results should not be interpreted as proof that offering more products causes churn. Product suitability and the quality of the customer relationship should be investigated further.

4.10 Customer Activity Analysis

Table 4.9: Churn Rate by Active Member Status

Customer status	Customers	Exited	Retained	Churn rate
Inactive members	4,849	1,302	3,547	26.85%
Active members	5,151	735	4,416	14.27%

Figure 4.9: Customer Churn Rate by Account Activity



Interpretation:

Inactive customers have a churn rate of 26.85%, compared with 14.27% for active customers. The difference is 12.58 percentage points.

This is one of the most important findings of the study. Customers classified as inactive are substantially more likely to have exited than active customers in the dataset.

The bank could monitor declining customer engagement, reduced use of banking services, and inactivity indicators. Appropriate engagement initiatives may include service reminders, customer feedback requests, product reviews, and assistance with relevant banking services.

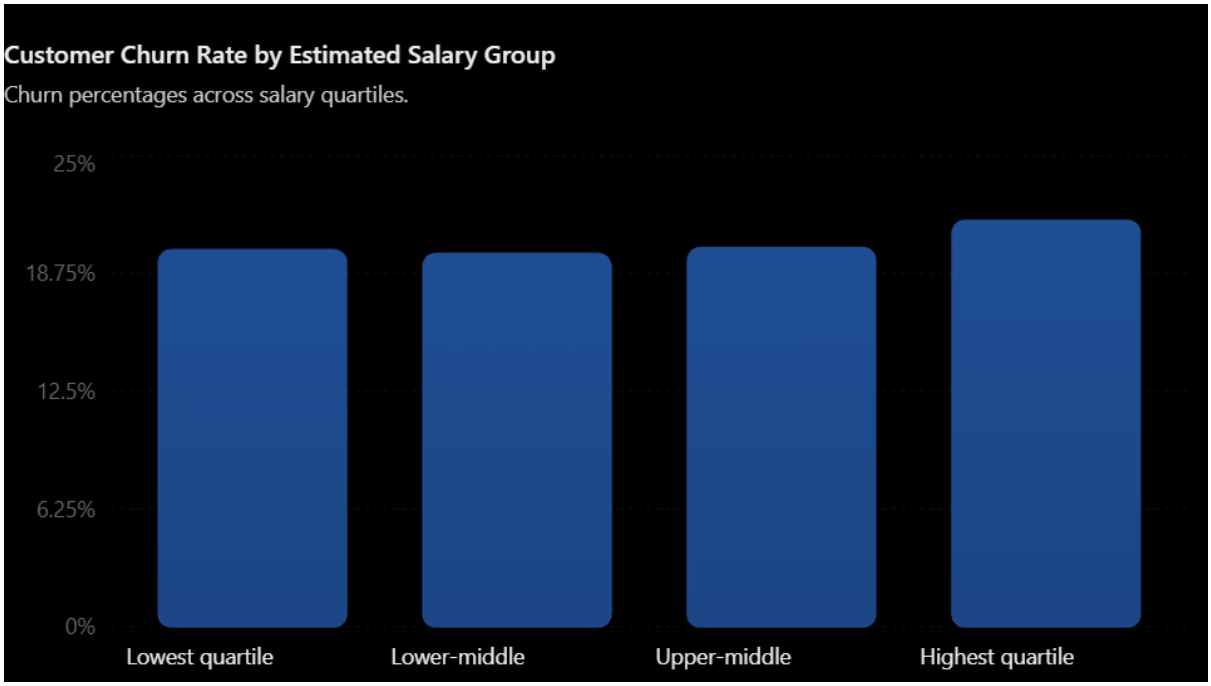
However, this comparison does not establish that inactivity directly causes churn. It identifies a useful pattern for further investigation.

4.11 Salary-Based Customer Churn Analysis

Table 4.10: Churn Rate by Estimated Salary Group

Salary group	Customers	Exited	Retained	Churn rate
Lowest quartile	2,500	500	2,000	20.00%
Second quartile	2,500	495	2,005	19.80%
Third quartile	2,500	503	1,997	20.12%
Highest quartile	2,500	539	1,961	21.56%

Figure 4.10: Customer Churn Rate by Estimated Salary Group



Interpretation:

The churn rate across salary groups ranges from 19.80% to 21.56%. The highest salary group has the highest churn rate at 21.56%, while the second salary group has the lowest at 19.80%.

The differences are relatively small. Estimated salary does not show a strong separation between high- and low-churn groups in this descriptive analysis.

The bank should avoid relying on salary alone to identify customers at risk of leaving. Other characteristics, including age, geography, product usage, and activity status, may provide more useful information when examined together.

4.12 Comparison of Retained and Exited Customers

Table 4.11: Average Characteristics of Retained and Exited Customers

Variable	Retained customers	Exited customers
Average age	37.41 years	44.84 years
Average credit score	651.85	645.35
Average tenure	5.03 years	4.93 years
Average account balance	72,745.30	91,108.54
Average number of products	1.54	1.48
Credit card ownership indicator, average	0.707	0.699
Active member indicator, average	0.555	0.361
Average estimated salary	99,738.39	101,465.70

Interpretation:

The comparison shows several differences between exited and retained customers.

First, exited customers are older on average. Their mean age is approximately 44.84 years, compared with 37.41 years among retained customers. This is consistent with the higher churn rates observed in the older age groups.

Second, exited customers have a higher average account balance. The average balance of exited customers is approximately 91,108.54, compared with 72,745.30 among retained customers. This does not mean that a higher balance causes churn; it indicates a difference between the two groups.

Third, the average activity indicator is lower among exited customers. The mean indicator is approximately 0.361 for exited customers and 0.555 for retained customers. This supports the earlier finding that inactive customers have a higher observed churn rate.

The average credit scores, tenure, product counts, credit card ownership indicators, and estimated salaries differ less substantially than age, account balance, and activity status.

Overall, the comparison suggests that age, engagement, and certain financial characteristics deserve further attention when investigating customer churn.

4.13 Year-Wise Analysis

The dataset contains 10,000 customer records associated with the year 2025.

Table 4.12: Year-Wise Churn Summary

Year	Customers	Exited	Churn rate
2025	**10,000**	**2,037**	**20.37%**

Interpretation:

The available data records a churn rate of 20.37% for 2025. Because the dataset contains no other year, it is not possible to compare churn across different years or determine whether churn is increasing or decreasing over time.

Future research could use datasets covering several years to examine changes in customer retention, seasonal patterns, and the effectiveness of retention initiatives.

4.14 Summary of Major Findings

The major findings of the analysis are summarized below.

1. The dataset contains 10,000 customer records, with an overall churn rate of 20.37%.
2. Germany has the highest geographical churn rate at 32.44%.

3. Female customers have a higher observed churn rate than male customers in the dataset.
4. Customers aged 50 and above have the highest age-group churn rate at 45.45%.
5. Credit score groups show relatively limited and non-linear differences in churn.
6. Tenure groups have churn rates between 19.67% and 21.15%.
7. Customers with positive balances have higher observed churn rates than customers with zero balances, although the pattern varies across balance groups.
8. Customers holding three or four products have particularly high observed churn rates, but these groups are relatively small.
9. Inactive customers have a churn rate of 26.85%, compared with 14.27% for active customers.
10. Salary-group churn rates are relatively similar, ranging from 19.80% to 21.56%.
11. Exited customers are older on average and have higher average account balances than retained customers.
12. The single-year dataset does not support year-over-year trend analysis.

4.15 Conclusion

The analysis identifies meaningful differences in churn across several customer segments. Geography, age, product count, and activity status show particularly notable differences in the observed churn rates.

Germany, customers aged 50 and above, inactive customers, and customers holding three or four products warrant further investigation. However, the findings are descriptive and should be validated before being used to make customer-level decisions.

The results provide a foundation for developing customer retention recommendations. The next chapter presents practical suggestions, the overall conclusion, study limitations, and possible directions for future research.

4.16 Identification of High-Churn Customer Segments

The customer segmentation analysis helps identify groups with comparatively higher observed churn rates. This comparison is useful for understanding where further customer-retention investigation may be required.

Table 4.13: Summary of Customer Segments with Higher Observed Churn Rates

Customer Segment	Total Customers	Exited Customers	Churn Rate
Four banking products	60	60	100.00%
Three banking products	266	220	82.71%
Age 50 and above	1,395	634	45.45%
Germany	2,509	814	32.44%
Age 40–49	2,618	806	30.79%
Inactive customers	4,849	1,302	26.85%
Female customers	4,543	1,139	25.07%

Interpretation:

The comparison shows that the highest observed churn rates occur among customers holding four products and customers holding three products. However, these groups contain only 60 and 266 customers respectively. Therefore, their high churn rates should be interpreted cautiously and should not be treated as conclusive evidence about all customers with multiple products.

Among the larger customer groups, customers aged 50 and above record a churn rate of 45.45%, while Germany records a churn rate of 32.44%. Customers aged 40–49 and inactive customers also demonstrate comparatively high churn rates.

These results indicate that age, geography, activity status, and product count are important areas for further investigation. The bank should examine additional customer information before deciding on retention actions.

4.17 Comparative Interpretation of Major Customer Segments

The analysis identifies differences in churn across geographical, demographic, financial, and behavioural customer groups.

Geographical analysis shows that Germany has a churn rate of 32.44%, compared with 16.15% in France and 16.67% in Spain. This indicates a considerable difference in the observed churn patterns across the three geographical categories.

Age-based analysis shows that customers aged 50 and above have the highest churn rate among the age groups, at 45.45%. The 40–49 age group also records a relatively high churn rate of 30.79%. In comparison, customers below 30 have a churn rate of 7.56%.

Customer engagement is another important area. Inactive customers have a churn rate of 26.85%, while active customers have a rate of 14.27%. This difference suggests that activity status is useful for understanding customer-retention patterns.

Product usage analysis shows that customers holding two products have a lower observed churn rate of 7.58% than customers holding one product, at 27.71%. The results for customers holding three and four products are unusually high, but these groups have smaller sample sizes.

Credit score, tenure, and estimated salary groups show comparatively limited differences in churn rates. Account balance also does not show a consistent increasing or decreasing pattern.

Overall, the analysis indicates that customer churn should be examined through multiple characteristics rather than relying on a single variable.

4.18 Limitations in Interpreting High-Churn Segments

Although the analysis identifies customer groups with higher observed churn rates, these results should be interpreted carefully.

The dataset provides information about customer characteristics and exit status but does not directly explain why customers leave the bank. Factors such as customer satisfaction, service quality, complaint history, transaction frequency, competitor offers, and reasons for account closure are not available in the examined variables.

The product groups containing three and four products also have relatively small sample sizes. Their churn rates may therefore be sensitive to the limited number of records.

The results represent descriptive comparisons and do not establish cause-and-effect relationships. Further investigation using additional customer information would be necessary before implementing customer-level retention decisions.

CHAPTER V

FINDINGS, SUGGESTIONS AND CONCLUSION

5.1 Introduction

This chapter presents the major findings of the study, practical recommendations for improving customer retention, the overall conclusion, limitations, and scope for future research.

The study examined 10,000 customer records from a European banking dataset to understand how customer characteristics are associated with observed churn patterns. The findings provide useful descriptive insights that can help a bank identify segments requiring further investigation and design more focused retention initiatives.

5.2 Major Findings of the Study

5.2.1 Overall Churn

The overall customer churn rate is 20.37%. Out of 10,000 customers, 2,037 exited and 7,963 remained.

This indicates that customer attrition is an important area for management attention.

5.2.2 Geographical Differences

Germany records the highest churn rate at 32.44%, while France and Spain record rates of 16.15% and 16.67%, respectively.

This indicates a substantial regional difference in observed churn. The bank should investigate customer experience and service-related factors that might explain the difference.

5.2.3 Age-Related Differences

Customers aged 50 and above have the highest churn rate at 45.45%, followed by customers aged 40–49 at 30.79%.

Younger customers have lower observed churn rates, particularly customers under 30, whose rate is 7.56%.

The findings indicate that age groups differ considerably in their churn patterns.

5.2.4 Gender Differences

Female customers have a churn rate of 25.07%, compared with 16.46% among male customers.

This difference warrants further investigation into customer needs, service experiences, and other characteristics. Gender should not be treated as a cause of churn.

5.2.5 Product Usage

Customers holding one banking product have a churn rate of 27.71%, while customers holding two products have a churn rate of 7.58%.

Customers holding three and four products have much higher observed churn rates. However, these groups contain relatively few customers, particularly the group holding four products.

The results suggest that product count and product combinations should be investigated further rather than assuming that more products automatically improve retention.

5.2.6 Customer Activity

Inactive customers have a churn rate of 26.85%, compared with 14.27% for active customers.

This difference makes customer engagement an important area for further analysis and potential retention initiatives.

5.2.7 Account Balance

Customers with zero balances have a churn rate of 13.82%, while positive-balance groups have higher rates ranging from 20.61% to 27.38%.

The relationship between balance and churn is not consistently increasing. Account balance should therefore be evaluated together with other customer characteristics.

5.2.8 Credit Score and Tenure

Credit-score groups show no clear, consistent trend in churn. Tenure-group churn rates are also relatively similar, ranging from 19.67% to 21.15%.

These variables may provide useful context, but they do not separate customer groups as strongly as some other characteristics in this descriptive analysis.

5.2.9 Estimated Salary

Salary-group churn rates range from 19.80% to 21.56%. The observed differences are comparatively small, suggesting that salary alone is not a strong segment discriminator in these results.

5.2.10 Retained and Exited Customer Differences

Exited customers have a higher average age and a higher average account balance than retained customers. They also have a lower average activity indicator.

These differences reinforce the importance of examining demographic characteristics, engagement, and financial variables together.

5.3 Suggestions and Recommendations

Based on the observed patterns, the following recommendations are proposed.

5.3.1 Investigate High-Churn Geographical Regions

The bank should prioritize an investigation of Germany, where the observed churn rate is 32.44%.

Recommended actions include:

- Review customer feedback and complaints by region.
- Examine differences in customer service response times.
- Compare product suitability, fees, and customer experience across regions.
- Conduct customer interviews or surveys where appropriate.
- Evaluate the effectiveness of region-specific retention initiatives.

The purpose should be to identify the reasons behind the regional difference before introducing targeted interventions.

5.3.2 Develop Appropriate Retention Support for Older Customers

Customers aged 50 and above have a churn rate of 45.45%, making this group a priority for further research.

The bank could:

- Review whether products meet the needs of this age group.
- Offer clear information about available banking services and fees.
- Provide accessible customer support.

- Collect feedback about customer satisfaction and service difficulties.
- Review whether customer needs differ across age groups.

These actions should be based on customer preferences and service needs rather than assumptions about older customers.

5.3.3 Improve Customer Engagement

Inactive customers have a higher observed churn rate than active customers.

The bank could develop an engagement program that includes:

- Monitoring periods of reduced account activity.
- Sending appropriate reminders about available services.
- Providing information about digital banking facilities.
- Requesting feedback from customers whose engagement has declined.
- Offering assistance when customers experience difficulties using banking services.

Communication should be relevant, respectful, and consistent with customer consent and applicable privacy requirements.

5.3.4 Review Product Usage and Product Suitability

The analysis shows substantial differences in churn according to the number of products held.

The bank should review the product combinations associated with higher churn, especially among customers holding three or four products. It should also examine whether customers holding one product have unmet needs or insufficient awareness of other suitable services.

Cross-selling should be based on customer requirements and product suitability. The bank should not assume that increasing the number of products will automatically prevent churn.

5.3.5 Investigate Customer Experience and Satisfaction

The available dataset does not contain direct measures of customer satisfaction, service quality, or reasons for exit.

The bank should consider collecting information about:

- Customer satisfaction.

- Complaint frequency and resolution.
- Service accessibility.
- Digital banking experience.
- Product suitability.
- Reasons for account closure or discontinuation.

This information would help explain the patterns identified in the descriptive analysis.

5.3.6 Use Multiple Variables for Customer Risk Assessment

The results indicate that churn patterns differ across several characteristics. Therefore, the bank should avoid relying on a single variable when deciding which customers may require additional support.

A future analytical model could consider age, geography, activity status, product count, tenure, account balance, credit score, and other relevant information.

Such a model should be evaluated using appropriate validation techniques before being used operationally.

5.3.7 Monitor Retention Performance

The bank should establish regular reporting on key indicators such as:

- Overall customer churn rate.
- Customer retention rate.
- Churn rate by geography.
- Churn rate by age group.
- Churn rate by activity status.
- Churn rate by product group.
- Customer satisfaction and complaint resolution.
- Retention following customer support or engagement initiatives.

Monitoring these indicators over time would allow the bank to determine whether retention initiatives are effective.

5.3.8 Maintain Responsible Use of Customer Data

Customer data should be handled securely and used only for appropriate purposes. Customer identifiers should be excluded from reports where they are not necessary.

Any future churn-risk system should be reviewed for data quality, privacy, fairness, and the potential for inappropriate customer treatment.

5.3.9 Segment-Based Retention Prioritization

The findings suggest that banks should organize retention investigations according to the observed churn patterns of different customer segments.

The following groups may be considered for further review:

1. **Geographical segment:** Germany records a churn rate of 32.44%. The bank should investigate regional differences in customer experience, service quality, and product suitability.
2. **Older customer segment:** Customers aged 50 and above record a churn rate of 45.45%. Their banking needs, service expectations, and customer feedback should be examined.
3. **Inactive customer segment:** Inactive customers have a churn rate of 26.85%. Monitoring declining activity and providing relevant customer assistance may support engagement.
4. **Product usage segment:** Customers holding three or four products demonstrate unusually high observed churn rates. The bank should investigate product combinations and suitability before making decisions.
5. **Gender segment:** Female customers have a higher observed churn rate than male customers in this dataset. Further investigation should consider customer needs and other relevant characteristics rather than assuming gender is the reason for churn.

The prioritization should be treated as a guide for further investigation, not as a prediction that an individual customer will leave.

5.3.10 Establish a Regular Customer Retention Review

Banks can improve their understanding of customer churn by reviewing important indicators regularly.

A customer-retention review may include:

- Total number of customers.
- Number of exited and retained customers.
- Overall churn and retention rates.
- Country-wise churn rates.
- Age-group churn rates.
- Active and inactive customer comparisons.
- Product-count analysis.
- Customer feedback and complaint information.
- Results of retention initiatives.

Regular reporting can help management identify changes in customer behaviour when data from multiple reporting periods becomes available.

5.4 Expected Benefits of the Recommendations

The recommendations provided in this study may support banking organizations in the following ways:

Improved customer understanding: Segmentation helps banks understand differences among customer groups.

Focused retention activities: Management can investigate customer groups with comparatively higher observed churn rates.

Better customer engagement: Monitoring activity status may help identify customers who require additional communication or assistance.

Improved decision-making: Data-based reporting provides a structured view of customer churn and retention.

Responsible customer management: Retention decisions can be based on relevant customer needs and verified information rather than assumptions.

The benefits would need to be evaluated through actual implementation and measurement of retention outcomes.

5.5 Managerial Implications

The findings have several implications for banking management.

First, customer churn should be monitored at the segment level rather than only through an overall percentage. The overall churn rate of 20.37% provides a useful baseline, but it does not reveal the substantial differences among customer groups.

Second, management should investigate high-churn segments before allocating retention resources. The elevated churn rates in Germany, among customers aged 50 and above, and among inactive customers suggest areas for further review.

Third, management should distinguish between observed associations and proven causes. A higher churn rate in a customer segment does not necessarily mean that the segment's characteristics caused the exit.

Fourth, the bank could improve decision-making by combining descriptive analysis with additional customer feedback and, in future work, a validated predictive model.

Finally, retention initiatives should be evaluated using measurable outcomes. The bank should compare churn rates before and after an intervention while accounting for other factors that may influence the results.

5.6 Limitations of the Study

The study has the following limitations:

1. **Dataset coverage:** The analysis is based on the available 10,000 customer records and may not represent the entire European banking industry.
2. **Single-year data:** The dataset contains records for 2025 only, preventing meaningful year-over-year comparisons.
3. **Descriptive analysis:** The analysis identifies differences and patterns but does not establish causation.

4. **Limited customer-experience information:** Variables such as satisfaction, complaints, service quality, and reasons for exit are unavailable.
5. **Small product groups:** The groups with three and four products contain relatively few customers, requiring caution when interpreting their churn rates.
6. **No predictive validation:** The study does not currently include a trained and validated model for forecasting future churn.
7. **Potential unobserved factors:** External influences and other customer characteristics not included in the dataset may affect churn.
8. **Grouping decisions:** The age, tenure, balance, credit-score, and salary groupings influence how the results are presented.

These limitations should be considered when interpreting the findings and applying the recommendations.

5.7 Scope for Future Research

Future studies could extend the present analysis in several ways.

5.7.1 Multi-Year Churn Analysis

Data covering several years could be used to study changes in churn, customer retention trends, and the impact of business initiatives over time.

5.7.2 Predictive Churn Modelling

Machine-learning methods such as logistic regression, decision trees, random forests, or gradient-boosting models could be explored to estimate customer churn risk.

The models should be evaluated using appropriate train-test separation, class-imbalance handling where necessary, and metrics such as precision, recall, F1-score, ROC-AUC, and precision-recall performance.

5.7.3 Customer Satisfaction Analysis

Future research could combine customer survey responses and service feedback with account data to understand the reasons behind churn.

5.7.4 Advanced Customer Segmentation

Clustering methods could be used to identify customer groups with similar financial characteristics, engagement patterns, and product usage.

5.7.5 Retention Intervention Evaluation

A future study could assess whether specific customer support or engagement initiatives are associated with improved retention, using an appropriate comparison design.

5.7.6 Explainable Analytics

Future work could investigate interpretable models and explainability techniques to help banking managers understand which factors contribute most to a model's churn-risk estimates.

5.8 Conclusion

The study, titled “Customer Segmentation & Churn Pattern Analytics in European Banking,” examined 10,000 customer records to understand observed churn patterns across demographic, geographical, financial, and engagement-related variables.

The overall churn rate was 20.37%, with 2,037 exited customers and 7,963 retained customers. The analysis identified higher churn rates in Germany, among customers aged 50 and above, among inactive customers, and among customers holding three or four banking products.

The results also showed that credit score, tenure, and estimated salary groups had comparatively limited differences in churn rates. The observed relationship between account balance and churn was not consistently increasing.

The study demonstrates how Python-based data analytics can help organize banking customer information, compare customer segments, identify areas requiring further investigation, and support evidence-informed retention planning.

The findings should be treated as descriptive evidence rather than proof of causation. Further research using customer-experience data, multi-year records, and validated predictive models could provide deeper insight into the reasons behind churn and help banks develop more effective retention strategies.

In conclusion, customer segmentation and churn pattern analysis can support better-informed banking decisions by highlighting differences among customer groups and identifying opportunities for further investigation, improved customer engagement, and responsible retention management.

CHAPTER 6

CUSTOMER CHURN DASHBOARD DEVELOPMENT

6.1 Introduction

Data analysis produces useful findings, but presenting the results in a clear and accessible format is equally important. A dashboard brings key indicators, charts, and filters together so that users can explore customer information without manually reviewing multiple tables.

This chapter describes the design and proposed implementation of a Customer Segmentation and Churn Analytics Dashboard for the European banking dataset. The dashboard is intended to provide an interactive view of customer churn, retention, geographical differences, demographic patterns, and customer engagement.

The dashboard can be developed using Python, Pandas, Matplotlib, and Streamlit. Its final contents and testing results should be updated to match the version actually implemented.

6.2 Objectives of the Dashboard

The main objectives are:

1. To display the total number of customers and overall churn rate.
2. To compare retained and exited customers.
3. To present churn rates across geographical regions.
4. To explore customer churn by age, gender, and tenure.
5. To examine churn in relation to product count and activity status.
6. To compare churn across financial-characteristic groups.
7. To allow users to filter customer segments.
8. To communicate findings in a user-friendly format.

6.3 Tools and Technologies

Tool or Technology	**Purpose**
Python	Main programming language
Pandas	Data loading, cleaning, grouping, and analysis

NumPy	Numerical operations
Matplotlib	Charts and data visualization
Streamlit	Interactive dashboard interface
Excel	Reviewing and storing analysis results
Google Colab or a local Python environment	Running analysis and testing code

These tools provide a practical way to convert the analytical results into an interactive presentation.

6.4 Dataset Description

The dashboard is based on the European banking dataset containing 10,000 customer records and 14 columns.

The dataset includes customer characteristics such as:

- Geography
- Gender
- Age
- Credit score
- Tenure
- Account balance
- Number of products
- Credit card ownership
- Active-member status
- Estimated salary
- Customer exit status

The available year field contains 2025 only. Therefore, the dashboard should focus on customer-level comparisons rather than year-over-year trends.

6.5 Proposed Dashboard Architecture

The dashboard follows a simple data-processing structure.

Step 1: Data input

The application reads the customer dataset from a CSV file.

Step 2: Data preparation

The application checks column names, data types, missing values, and the exit indicator. It creates customer groups required for analysis.

Step 3: KPI calculation

The application calculates customer count, exited customers, retained customers, churn rate, and retention rate.

Step 4: Visualization

Charts display churn patterns across geography, age groups, gender, tenure, activity status, product count, and financial groups.

Step 5: User interaction

Users select filters to examine the segments relevant to their analysis.

Step 6: Interpretation

The displayed results help users identify differences between customer segments and consider possible retention actions.

6.6 Dashboard Layout

The proposed dashboard is organized into the following sections.

6.6.1 Dashboard Title and Introduction

The top of the dashboard displays the title:

Customer Segmentation & Churn Pattern Analytics in European Banking

A short description explains that the dashboard provides an overview of customer churn patterns and segment-level comparisons.

6.6.2 Key Performance Indicators

The dashboard should display the following KPIs based on the full dataset:

KPI	Value
Total Customers	10,000
Exited Customers	2,037
Retained Customers	7,963
Churn Rate	20.37%
Retention Rate	79.63%

These values represent the overall dataset. If filters are applied, the displayed values should update to represent the selected customer subset.

6.6.3 Customer Churn Distribution

A pie chart or bar chart displays the proportion of customers who exited and remained.

The overall churn rate is 20.37%, while the retention rate is 79.63%. This visualization provides an immediate summary of the customer base.

6.6.4 Geography-Based Analysis

A bar chart compares churn rates across France, Germany, and Spain.

The observed rates are:

Geography	Churn Rate
France	16.15%
Germany	32.44%
Spain	16.67%

Germany demonstrates the highest observed churn rate in this dataset. This result can guide further investigation into regional customer experience and service conditions.

6.6.5 Age-Based Analysis

An age-group chart compares churn rates for customers under 30, aged 30–39, aged 40–49, and aged 50 and above.

The observed churn rates are 7.56%, 10.88%, 30.79%, and 45.45%, respectively.

The chart highlights higher observed churn among older customer groups. These results describe the dataset and should not be interpreted as evidence that age itself causes churn.

6.6.6 Gender-Based Analysis

A gender comparison chart displays churn rates of 25.07% for female customers and 16.46% for male customers.

The dashboard should present these differences neutrally and encourage further investigation into service experience and customer needs rather than treating gender as a direct cause of churn.

6.6.7 Tenure-Based Analysis

A tenure chart compares customers with 0–2 years, 3–5 years, and six or more years of tenure.

The observed churn rates are 21.15%, 20.76%, and 19.67%, respectively.

The results indicate relatively small differences across tenure groups.

6.6.8 Activity-Based Analysis

The activity chart compares active and inactive customers.

Activity Status	Churn Rate
Active Members	14.27%
Inactive Members	26.85%

The dashboard highlights the higher observed churn rate among inactive customers. This may support further examination of customer engagement and retention initiatives.

6.6.9 Product-Based Analysis

A product-count chart compares churn rates among customers holding one, two, three, or four products.

The observed rates are 27.71%, 7.58%, 82.71%, and 100%, respectively.

The dashboard should display sample sizes alongside these rates. In particular, the four-product group contains only 60 customers, so the 100% churn rate should be interpreted cautiously. A high rate in a small group may be unstable and should not be treated as conclusive evidence.

6.6.10 Financial Characteristics

The dashboard may include charts for credit score, balance group, and estimated salary group.

These visualizations allow users to examine whether churn rates differ across the financial categories used in the analysis. They should be presented as descriptive comparisons rather than causal explanations.

6.7 Interactive Filters

The dashboard may include filters for:

- Geography
- Gender
- Age group
- Customer activity status
- Number of products
- Tenure group

When a user changes a filter, the customer count, churn rate, retention rate, and relevant charts should update to represent the selected records.

The application should clearly indicate when the displayed results refer to a filtered subset rather than the full dataset.

6.8 Data Processing and KPI Formulas

The churn and retention KPIs can be calculated as follows:

Churn Rate

Churn Rate = (Number of Exited Customers / Total Customers) × 100

Retention Rate

Retention Rate = (Number of Retained Customers / Total Customers) × 100

For the full dataset:

Churn Rate = (2,037 / 10,000) × 100 = 20.37%

Retention Rate = (7,963 / 10,000) × 100 = 79.63%

The two percentages sum to 100% because every customer is classified as either exited or retained in the analysis.

6.9 Dashboard Implementation Procedure

The following procedure can be followed to develop the dashboard.

1. Prepare the CSV dataset and confirm the file location.
2. Install Python dependencies, including Streamlit, Pandas, NumPy, and Matplotlib.
3. Create a Python file named app.py.
4. Import the required libraries.
5. Load the dataset and validate its columns.
6. Prepare the churn indicator and customer segment categories.
7. Calculate the overall and filtered KPIs.
8. Create the charts for customer distribution and segment-level churn.
9. Add interactive filters.
10. Run the application and inspect the dashboard.
11. Compare displayed results with the analysis workbook.
12. Capture screenshots of the final working dashboard for the report.

The final report should include screenshots only after these steps have actually been completed.

6.10 Dashboard Testing

Testing is necessary to verify that the dashboard presents correct results and responds appropriately to user selections.

Test Case	Expected Result
Load the CSV file	Dataset loads without errors
Check customer count	Full dataset shows 10,000 records
Check exited customers	Full dataset shows 2,037 exited customers
Check churn rate	Full dataset shows 20.37%
Apply a geography filter	KPIs reflect the selected geography
Apply an age filter	KPIs reflect the selected age group
Apply an activity filter	KPIs reflect the selected activity group
Check charts	Chart values agree with the underlying data
Check missing or invalid data	Application handles issues appropriately
Clear filters	Full-dataset results return

These are the proposed test cases. In the final submission, mark each test as passed or failed based on actual execution.

6.11 Dashboard Limitations

1. Dashboard accuracy depends on the quality of the underlying dataset.
2. The dataset covers only 2025, limiting time-based analysis.
3. Descriptive churn rates do not predict which individual customers will leave in the future.
4. Some customer segments have relatively small sample sizes.
5. The available variables may not explain the underlying reasons for customer exit.
6. The dashboard cannot establish causal relationships from simple comparisons.

7. Operational deployment would require appropriate access controls, data protection, and maintenance.

6.12 Future Enhancements

The dashboard could be improved by adding:

1. Validated churn prediction models.
2. Model-performance measures such as precision, recall, F1-score, and ROC-AUC where appropriate.
3. Customer-level risk scores, subject to appropriate validation and governance.
4. Additional information on customer complaints, transactions, satisfaction, and service interactions.
5. Time-based analysis if data from multiple periods becomes available.
6. Export functionality for selected tables and charts.
7. Improved accessibility and visual design.
8. Privacy and security controls for any real customer information.

6.13 Chapter Summary

This chapter describes the proposed development of a customer churn dashboard for the European banking dataset. The dashboard brings together key performance indicators, customer segmentation, churn-rate comparisons, visualizations, and interactive filters.

The dashboard is intended to make analytical findings easier to understand and explore. Its usefulness depends on correct implementation, validation of the displayed results, and responsible interpretation of customer data.

6.14 Key Dashboard Performance Indicators

The dashboard is intended to provide a consolidated view of the customer base through important performance indicators.

Table 6.1: Dashboard KPI Summary

Key Performance Indicator	Dataset Value

Total Customers	10,000
Exited Customers	2,037
Retained Customers	7,963
Overall Churn Rate	20.37%
Overall Retention Rate	79.63%

These values represent the complete dataset before applying any filters. When filters are applied, the KPI values should be recalculated for the selected customer group.

6.15 Dashboard Interpretation

The dashboard is intended to help users understand the overall customer churn position and compare different customer segments.

The overall churn rate of 20.37% provides a baseline for evaluating customer attrition. The geographical comparison highlights Germany as the category with the highest observed churn rate. The age-based comparison highlights customers aged 50 and above, while the activity analysis shows a higher churn rate among inactive customers.

The product-count comparison also highlights differences between customers holding one, two, three, and four products. The results for the three-product and four-product groups should be interpreted carefully because their sample sizes are comparatively small.

By combining these indicators in one interface, the dashboard can make the analytical results easier to review and communicate.

6.16 Dashboard Validation Approach

Dashboard validation is important to ensure that the displayed figures are consistent with the underlying dataset and the analysis workbook.

The following checks should be performed:

1. Confirm that the dataset loads correctly.
2. Verify that the full dataset contains 10,000 customer records.
3. Confirm that the number of exited customers is 2,037.

4. Confirm that the number of retained customers is 7,963.
5. Verify that the overall churn rate is 20.37%.
6. Apply geographical filters and compare the displayed results with the analysis workbook.
7. Apply age and activity filters and verify the recalculated KPIs.
8. Check whether the charts update correctly after filter selection.
9. Clear the filters and confirm that the full-dataset results return.

The testing results should be recorded only after the dashboard has actually been executed. Screenshots and test evidence should be included in the final report to demonstrate the completed work.

6.17 Dashboard Limitations and Future Improvements

The dashboard is based on a dataset containing customer records for 2025 only. Therefore, it cannot show year-over-year changes in customer churn.

The dashboard presents descriptive churn information and does not independently predict future customer exits. Its usefulness also depends on the accuracy of the underlying data and the correct calculation of customer-segment indicators.

Future improvements may include:

- Adding multi-year customer information.
- Including customer complaints and satisfaction indicators.
- Incorporating transaction and digital-banking activity.
- Developing a validated customer churn prediction model.
- Adding export options for selected charts and tables.
- Improving privacy and access controls before using actual customer information.

These improvements could make the dashboard more useful for ongoing customer-retention analysis.

REFERENCES

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APPENDIX

Appendix A: Dataset Variables and Their Purpose

Variable	Description	Purpose in the Study
CustomerId	Unique customer identifier	Identifying customer records
Year	Year associated with the record	Identifying the available data period
Geography	Customer geographical category	Geographical segmentation
Gender	Customer gender category	Gender-based analysis
Age	Customer age	Age-based segmentation
CreditScore	Customer credit score	Credit-score analysis
Tenure	Length of the banking relationship	Tenure-based analysis
Balance	Customer account balance	Financial segmentation
NumOfProducts	Number of banking products held	Product usage analysis
HasCrCard	Credit card ownership indicator	Credit card ownership analysis
IsActiveMember	Customer activity indicator	Customer engagement analysis
EstimatedSalary	Estimated customer salary	Salary-based analysis
Exited	Customer exit indicator	Measuring customer churn
Surname	Customer surname	Dataset identification field; not used as an analytical factor

The Exited variable is the main outcome variable used in the analysis. A value of 1 represents an exited customer, while a value of 0 represents a retained customer.

CustomerId and Surname are identification-related fields and are not used to explain customer churn patterns.

Appendix B: Data Analysis Outputs

The following outputs should be included as supporting evidence:

1. Dataset dimensions and column information.
2. Missing-value assessment.
3. Duplicate-record assessment.
4. Overall customer churn summary.
5. Country-wise churn analysis.
6. Age-wise churn analysis.
7. Gender-wise churn analysis.
8. Credit-score analysis.
9. Tenure analysis.
10. Account-balance analysis.
11. Product-count analysis.
12. Customer activity analysis.
13. Estimated-salary analysis.
14. Retained-versus-exited customer comparison.

Appendix C: Python Code and Project Evidence

The Python code used for the project should be included in the appendix. The code should cover:

- Dataset importation.
- Data cleaning and validation.

- Missing-value and duplicate checks.
- Creation of customer segments.
- Calculation of churn and retention rates.
- Group-wise customer analysis.
- Generation of charts.
- Exporting analytical results to Excel.

The final dashboard source code should also be included if the Streamlit application has been implemented.